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Aphorisms and Public Opinion: How the Socialization of Life Lessons Shapes Political Attitudes

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One of the fundamental problems of survey research is how best to measure latent attitudes. Many common measures are vague, easily primed, or endogenous to outcome variables of interest—or some combination therein. Rather than ask respondents to generalize via abstract measures, many of which they have never encountered or pondered, I introduce a new approach: measuring latent attitudes via orientations toward common aphorisms, e.g., "Don't judge a book by its cover," that express a worldview or ethical principle in a familiar yet pithy, axiomatic statement. Such sayings are socialized during the critical period of language development and are building blocks of language akin to idioms. Drawing on meme theory, I argue that aphorisms are readily transmittable forms of cultural information—memes—that are uniquely suited to minimize common survey measurement problems such as endogeneity and ambiguity. I apply this theory to risk attitudes, finding that aphorisms that tap risk acceptance, e.g., "Nothing ventured, nothing gained," constitute a highly coherent scale, are associated with known correlates of risk acceptance, and predict greater political participation. In doing so, I show that using attitudes toward common aphorisms presents a new method by which survey researchers can tap latent attitudes that are 1) theoretically exogenous to most political outcomes of interest and 2) concrete representations of abstract worldviews in the minds of respondents.

> *"Besides, it happens (how, I cannot tell) that an idea launched like a javelin in proverbial form strikes with sharper point on the hearer's mind and leaves implanted barbs for meditation..."*
>
> —Desiderius Erasmus, *Adages*

One's early-in-life experiences have an outsized impact on later-in-life political attitudes (Campbell et al. 1960; Green et al. 2004). For example, political events that happen during child and adolescent development can shape how cohorts of voters form political attachments (Valentino and Sears 1998) and engage with political rhetoric (Schuman and Corning 2006). The theory that socialized attitudes—beliefs about the world gained from experience during one's formative years—impact how individuals engage with politics and inform their resulting opinions is not a novel one (Sapiro 2004). Indeed, studies of parent-to-child (Jennings et al. 2009), child-to-parent (Linimon and Joslyn 2002; Dahlgaard 2018), peer-to-peer (Strother et al. 2021), and sibling-to-sibling (Healy and Malhotra 2013;

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Urbatsch 2011) socialization show the effects of early-in-life experience in shaping one's politics. While evidence of socialized attitudes' import abounds, the specific units of cultural and social information that are transmitted from one group or person to the child remain only moderately theorized and empirically untested at a generalizable, systematic level.

Moreover, what political socialization scholarship often neglects is how the *non-political* experiences and life lessons learned during childhood socialization shape one's outlook on the world, including when it comes to politics (Hyman 1959). Indeed, little work theorizes where these attitudes might originate in the socialization process and *how* they might be passed along, much less empirically tests the steps in the socialization process.[1] Yet early-in-life experiences are crucial for one's understanding of the world, including learned behaviors (Bandura 1977), attachments (Bowlby 1973), and social relationships (Jin and Yang 2026; Kanakogi et al. 2025) rather than societal, structural factors (Lansford et al. 2025).

However, such experiences are often varied and personalized, making measurement difficult, particularly after the period of socialization concludes. How might we measure the transmission of worldviews, non-political in nature but often political in application, that shape one's outlook?

A fruitful area for research in this area is the study of aphorisms. An aphorism is a concise distillation of a life lesson or worldview contained in the form of a memorable phrase. Aphorisms often have a ring of wisdom to them that makes them seem true axiomatically. When a child first hears the saying "Don't judge a book by its cover," it is likely conveyed with some conviction. Moreover, one intuitively knows that the meaning conveyed here is not specific to simply disregarding books with unattractive jackets. Take another example: "Strike while the iron is hot." One does not need to know anything about blacksmithing or the origin of the aphorism itself to comprehend or employ this aphorism. The meaning is intuitively clear (Cutting and Bock 1997), and the aphorism, by its metaphorical nature, is versatile enough to apply to many different situations (Lakoff and Johnson 1980).[2] Such sayings are building blocks of human language and are derived from our experiences

[1]However, see Hetherington et al. 2026, Sirin et al. 2021, Hatemi and Ojeda 2021, and Healy and Malhotra 2013.

[2]Though I refer to these statements as aphorisms, this term is used to encompass adages, proverbs, and aphorisms alike. Each are stand-alone statements that describe an approach to or an outlook on life. Idioms, though similar in form but not in meaning, do not and are therefore excluded from this research. Thus, "aphorism" is used as

(Lakoff and Johnson 1980), develop at an early age (Cacciari and Levorato 1989), and are stored in long-term memory (Győrfi 2017; Papagno 2001). By interacting with these concrete sayings, or aphorisms, in multiple abstract situations, an imprint is left in the psyche, and they become a part of how one communicates (Lakoff 1993). The cognitive pathways formed during this socialization period are likely important later on in life, too. Thus, measuring orientations toward aphorisms is a useful way to tap the cognitive processes formed during these years.

Though aphorisms are rarely used in survey research, where they have been used, fruitful results have arisen (Wilson 2025; Hetherington et al. 2026; Lansford et al. 2025). In the sections that follow, I develop a theory of aphorisms as measurement tools, arguing that aphorisms are uniquely useful distillations of non-political worldviews that are crystallized during socialization, making them cognitively accessible in the minds of survey respondents. Measuring latent attitudes via aphorisms, then, is more likely to inoculate researchers from common measurement problems arising from inaccurate respondent self-reports, abstract survey items, and endogeneity. Because familiarity with non-political aphorisms almost certainly develops during non-political socialization, attitudes toward them are theoretically prior, or exogenous, to any substantive understanding of politics, whether developed during the primary periods of socialization or thereafter. By tapping aphoristic attitudes in surveys, researchers can create survey items that are more "concrete" and that represent respondents' latent attitudes more accurately.

SOCIALIZED ATTITUDES

The period of pre-adult socialization is highly relevant to how citizens tend to form political opinions and identities (Campbell et al. 1960; Sapiro 2004). There are several routes through which political socialization takes place. Scholars generally focus on two areas: Parents can directly transmit political orientations, identities, and values, or they can serve as political role models for their children (Hyman 1959). Though initial work on the socialization of partisanship showed low rates of parent-child transmission (Jennings and Niemi 1968), later research has placed particular emphasis on the role

an umbrella term for any axiomatic statement that describes a specific way of viewing the world that is both generalizable and familiar via repeated cultural transmission.

that cue-giving and political engagement play in increasing the effects of political socialization (Jennings et al. 2009; Anoll et al. 2025), as well as the reception to these cues (Hatemi and Ojeda 2021). Additionally, in forming political attachments, personal experiences with politics play a role in accelerating socialization over the course of campaigns (Sears and Valentino 1997), particularly when those who are learning about politics engage in political discussions (Valentino and Sears 1998).

What scholars often overlook, however, is another mechanism by which Hyman argues political socialization might take place: the socialization of values that are *non-political* by nature but are nonetheless directly *analogous* to politics (Hyman 1959). Though increasing attention is being paid to such mechanisms (e.g., Hetherington et al. 2026; Healy and Malhotra 2013), perhaps the most notable exception to this rule is the study of authoritarianism, or authority-mindedness, as a predisposition. Key (1961) was among the first to conjecture that socialized attitudes toward authority might influence the way people view politics. Several decades later, the ANES added a 4-item measure of authority-mindedness that captures respondents' child-rearing attitudes. Studies of authority-mindedness have yielded results that showing responses to these four questions predict many political attitudes (e.g., gay rights, civil liberties, and immigration; Hetherington and Weiler 2009; Hetherington and Suhay 2011) and non-political behavior (e.g., sports, beer, and music preferences; Hetherington and Weiler 2018).

Indeed, as political cleavages have turned from the more traditional economic cleavages (e.g., the size and scope of the federal government) to social cleavages (e.g., gender equality), authority-mindedness has been shown to be central to describing the divide between Republicans and Democrats (Hetherington and Weiler 2009, 2018). Even as new salient issues crop up, such as threats from terrorism post-9/11, people filter this through their exogenous authority predispositions (Engelhardt et al. 2021) to inform their political preferences (Hetherington and Suhay 2011). When it comes to aphorisms, previous work shows that respondents' orientation toward a single aphorism tapping authority-mindedness, "Children ought to be seen and not heard," is correlated with the ANES measure, is highly constraining in respondents' belief systems, and is highly relevant to explaining abstract political attitudes, such as democratic norms (Wilson 2025).

Beyond authority-mindedness, socialized attitudes toward hypercompetitiveness—"a psychological need to win at all costs"—predict support for the January 6 insurrection among Republicans, more so than strength of partisanship, Trump support, racial attitudes, conspiratorial thinking, or Social Dominance Orientation (Hetherington et al. 2026). Furthermore, other scholars have shown that socialized attitudes toward *groups* shape political opinions, including specific policies that favor in-groups (Conover 1984), which is largely in line with Social Identity Theory (SIT; Tajfel and Turner (1979)). In contrast to SIT, other scholars have shown that socialized attitudes within racial minority groups can help predict support for punitive policies toward in-group members (Jefferson 2023) and policies that favor both in-groups and out-groups (Sirin et al. 2021).

In sum, the literature on socialized attitudes shows that citizens do indeed rely on *non-political* outlooks on life to inform their political opinions in a wide variety of contexts. Still, the study of how non-political socialization shapes public opinion runs into obvious roadblocks: It is extremely difficult to measure every aspect of socialization that might be relevant to politics—there are many beyond what has been outlined here. However, what specific information is being passed along; what are the *mechanisms* by which this information is socialized; and how might scholars go about measuring the resulting outlooks on life that are formed during this period?

“MEMES"

While the values conveyed through socialization can often be politically relevant (e.g., Galston (2001)), scholars know less about the specific pieces of information—or “memes”—that are conveyed to children during this period, including to what extent non-political information can be applied to politics and what form that non-political information takes. This is inherently a difficult problem. It is neither practical to observe every aspect of the parent-child socialization process, nor would an Institutional Review Board approve it.

A useful way to consider the socialization process is to think about what packets of information might be passed along from parent to child or through the culture more generally. One approach to conceptualize the transmission of non-political worldviews via aphorisms is genetic transmission in evolutionary biology. In sexual reproduction, genes are transmitted from two parents, one male and one female, onto offspring, and those offspring, if they survive long enough, pass those genes along to future generations. From the natural selection perspective, evolution occurs because the genes that are most helpful to one's survival are passed along to future generations (Lewontin 1970; Endler 1986).[3] These crucial genes can spread relatively quickly through a population if they are more useful than other genes. For example, an eye, extremely useful to an organism's survival, can easily develop in about half a million years, a vanishingly small fraction of evolutionary time (Schwab 2018).

Evolutionary biologist Richard Dawkins has analogized genetic transmission to the transmission of cultural information throughout a population that is helpful to its survival (Dawkins 1976). Dawkins refers to these pieces of social information by coining the term "meme" (Dawkins 1976).[4] Genes are replicators of DNA (genetic information), and memes are replications, or imitations, of cultural information; indeed, the term "meme" was coined from the Greek root word for memory (Dawkins 1976). A gene has a better chance at survival if it helps the organism survive in its environment. Similarly, memes are better or worse at surviving depending on the cultural environment in which they are produced and replicated.

When it comes to memes, however, we ourselves are the replicators. Therefore, memes are a product of human nature and our ability to produce and replicate social information. A meme is anything that includes the transmission of social information based on replication or imitation (Dawkins 1976). From how cooking pots are made to decorative art to revolutionary political manifestos (Dawkins 1976), memes need only be replicable to survive. Probably the most consequential meme according to Dawkins is religion, insofar as it was helpful to the survival of the human species (Dawkins 2006). As he points out, however, memetic transmission does not necessarily pertain to human survival, but rather to the survival of cultural information. Memes, like genes, can be spread quickly if they are particularly potent, but they can also lose steam quickly, like a fad, and need not have an evolutionary benefit (Dawkins 1976).

Another aspect of genetic transmission that is comparable to memetic transmission is copying-fidelity; the better the replicator is at transmitting the meme, the more likely the meme is to survive

[3] However, this is not the only perspective on the evolutionary process (e.g., Mayr 2001).

[4] Richard Dawkins was the original coiner of the term "meme." At the time, he was not referring to the common present-day understanding of a meme as an amusing combination of an image and text passed around on the Internet but rather anything that catches fire socially. Internet memes are a form of the meme Dawkins refers to, but they are a tiny subset of what he considers to be the memetic transmission of social information.

(Dawkins 1976). The copying-fidelity element of memetic survival is likely dependent on two important factors: the amount of information to be conveyed and the population's collective cognitive capacity to convey it.

For example, it is difficult to convey Karl Marx's views on religion succinctly, and long recitations of passages are unlikely to survive as a meme because it is too cognitively intensive for humans to do so on a mass scale. However, a specific quotation, such as "Religion is the opiate of the people," can suffice. Most have likely heard this quote from Marx and can impute his views about religion from it, even though it somewhat misrepresents his perspective. Much like all heuristics, it is not perfect, but it is still valuable as a shorthand for Marx's views on religion because it is pithy, succinct, and efficient.

For the purposes of this study, a specific *category* of memes—aphorisms—ought to be particularly useful in studying the socialization of non-political attitudes. Aphorisms contain underlying worldviews and convey one's approach to life in a uniquely concise fashion. Consider one of Aesop's fables: "The Tortoise and the Hare." A key lesson from this childhood story is that steadily working towards one's goal, not being flashy and careless, leads to success. A common aphorism derived from this story is "Slow and steady wins the race." The aphorism is itself both a heuristic for the moral of the story and a social meme that can be easily conveyed through childhood socialization. In that sense, aphorisms such as "Slow and steady wins the race" can be efficiently deployed to speak to how one approaches life, including how one thinks about politics. Take another example: the adage that one should "Fight fire with fire." The worldview inherent in this phrase was first introduced by William Shakespeare in *King John*, where he wrote "[B]e fire with fire." More recently, firefighters developed the tactic of literally fighting a large fire by lighting small ones around it. Regardless of where one traces its specific origin, the saying "Fight fire with fire" crops up when one feels beaten down, perhaps by an adversary, and decides to employ the same tactics to fight back, regardless of the destruction it might cause.[5]

In this sense, aphorisms are often drawn upon as a heuristic under philosophical conditions of uncertainty (i.e., according to what principles should one live one's life?), as other heuristics are (Tversky and Kahneman 1974). Aphorisms, like "Fight fire with fire," are useful heuristics for one's outlook on

[5]Examples abound of this particular turn of phrase being deployed in the current fight over partisan gerrymandering (e.g., Shorman 2025).

life because they effectively and efficiently communicate the underlying worldviews contained in them. In many cases, the aphorism is more valuable than the text or specific situation it references because aphorisms can be quickly and easily applied to a wide range of personal circumstances. Moreover, the attitudes toward aphorisms are not necessarily dependent on their original meaning, though this might enhance their effectiveness. One does not need to be familiar with the specifics of *King John* or firefighting techniques to understand the meaning of the expression "Fight fire with fire." Rather, aphorisms, modeled as memes, spread through a culture because of the mutual understanding of the philosophical punch they pack. Regardless of where one traces the etymology of the phrase, the meaning is intuitively understood; the phraseology is pithy; and the life lesson is practically generalizable, such that it is reasonable to assume most people who employ the saying do so in contexts that have nothing to do with fighting literal fires. Because aphorisms serve as a socialized heuristic in this way, they can be used by anyone familiar with the aphorism itself and the underlying worldview it conveys.

COGNITIVE ACCESSIBILITY

Aphorisms have untapped potential in survey research about politics.[6] A common criticism of scales used in survey research is whether respondents can accurately express their beliefs about the concept being measured (John 2008). As concepts become more abstract, this issue becomes increasingly thorny (Krosnick and Presser 2010). For example, consider personality trait scales. A key assumption in measuring personality traits through surveys is that of rationality, which means that respondents can accurately understand themselves (John 2008). However, small alterations in survey questions can result in vastly different responses to personality questions (Moskowitz 1986; Paulhus 1991), and self-reports of personality traits can often be biased based on the interpretations of survey questions (Kagan 2005). Indeed, relying on self-reports alone for personality-based research has come under heavy criticism (Kagan 1988) and has even been likened to biologists relying solely on similar self-reports to understand diseases (Kagan 2007). Many questions used in personality scales are simply too abstract to be useful

[6]However, see Wilson 2025; Hetherington et al. 2026, and Lansford et al. 2025.

and replicable.[7]

Due to their familiarity, aphorisms ought to be uniquely useful measurement tools because they are attitude-objects for which evaluations are readily accessible in the minds of survey respondents. Attitude accessibility is defined as the strength of the relationship between an attitude object and subsequent evaluation of that object (Fazio et al. 1982). In other words, the more easily retrievable an attitude is from one's memory, the more strongly one's associations are with the attitude object—in this case, aphorisms. In various contexts, cognitive accessibility has been shown to affect political attitudes (Fazio and Williams 1986; Houston and Fazio 1989; Taber and Lodge 2006). Moreover, the accessibility of one's political heuristics, such as partisanship and ideology, is predictive of how much one uses such heuristics when forming political opinions and judgments (Huckfeldt et al. 1999).

Though developed with a conceptually different goal in mind, Zaller (1992) provides a framework—the Receive, Accept, and Sample (RAS) model—for understanding why aphorisms ought to be accessible measurements of non-political socialization. According to the RAS model, citizens encounter political information (i.e., Receive) and choose whether to absorb that information or reject it (i.e., Accept) by filtering this information through their own experiences and predispositions (Zaller 1992). Because most survey respondents do not possess preconceived political attitudes, they tend to generate responses to survey questions on the spot (Zaller and Feldman 1992). In doing so, survey responses to abstract, or unfamiliar, political questions are largely a product of the considerations in the minds of survey respondents at the time they answer the survey question.

Aphorisms, on the other hand, ought to occupy a unique (and theoretically distinct) place in this model. Orientations toward aphorisms are primarily crystallized during the socialization process. One reason for this is that the transmission of aphorisms ought to be particularly memorable. The more a meme is replicated and transmitted through a population, the more familiar it is to respondents; and the more accurately an aphorism reflects one's worldview, the stronger the association between the attitude-object (i.e., an aphorism) and the attitude associated with the object (i.e., orientation toward the aphorism). Therefore, when survey respondents are asked to indicate their orientation toward an aphorism, they are much more likely to indeed contain a preconceived attitude. Thus, attitudes toward

[7] I describe an example of one such question measuring risk attitudes in a later section.

aphorisms are more likely to be accurate, exogenous representations of latent attitudes, rather than constructed by respondents when put on the spot in a survey questionnaire.

Though empirical public opinion research on aphorisms is scant, studies in cognitive linguistics suggest that formulaic language, such as idioms and proverbs, occupy a unique place in the mind that theoretically maps onto aphorisms (Honeck 1997; Gibbs and Beitel 1995). Formulaic language comprises approximately 25% of everyday speech (excluding filler words such as “um”; Van Lancker Sidtis and Sidtis 2018), and learning formulaic language is crucial for understanding the world and one’s place in it (Burdelski and Cook 2012). Such language, including aphorisms and proverbs, are relied upon as a cognitive shortcut to express oneself (Gibbs 1980; Huang et al. 2023). For example, children learn the meaning of idiomatic figurative language at a young age (Cacciari and Levorato 1989), and their structure is preserved in long-term memory, even among those suffering from aphasia or Alzheimer’s disease (Papagno 2001; Van Lancker Sidtis 2012; Győrfi 2017). Moreover, formulaic language sequences, such as idioms or aphorisms, are understood in the mind as a single conceptual unit, not as separate parts adding up to one whole (Cutting and Bock 1997), which allows for them to be deployed efficiently in communications (Gibbs 1980; Huang et al. 2023).

In sum, because the neural pathways forged during socialization, combined with the simple structure of the aphorism, allow for them to be easily accessible in cognition, aphorisms ought to be a particularly valuable measure of socialized beliefs. Measuring dispositions toward aphorisms, due to their cognitive accessibility, might also help explain more variation in dependent variables of interest, particularly when respondents are asked about abstract political concepts. Thus, predicting political opinions based on various latent attitudes might be significantly helped by using more concrete, cognitively accessible aphorisms rather than abstract self-reports.

RISK, POLITICAL PARTICIPATION, AND APHORISMS

To validate the use of aphorisms as a measurement tool, I turn to a previous study’s framework in analyzing the relationship between individuals’ risk acceptance and their willingness to participate in politics (Kam 2012). According to Kanthak and Woon (2015), “Risky decisions are those for which there is greater uncertainty—more specifically, greater variance—in the outcomes." In different contexts, attitudes toward risk have been shown to affect a broad range of political attitudes, such as Social Security privatization (Maestas and Pollock 2010) and free trade (Ehrlich and Maestas 2010). Moreover, risk attitudes can be primed in opinion questions (Eckles and Schaffner 2011), and policies framed in terms of risk result in different support depending on the individual respondent's disposition toward risk (Kam and Simas 2010). Additionally, there are demographic differences in individuals' risk attitudes: Women (Byrnes et al. 1999), younger individuals (Byrnes et al. 1999), and liberals (Kam 2012) are all more likely to accept risks.

Beyond public opinion, risk attitudes often affect political behaviors, including vote choice (Oshri et al. 2023) and strategic voting (Martin 2021). Additionally, when an individual decides whether to engage with the political process, they are weighing considerations that inherently contain uncertainty because the likelihood of effecting change through political participation is rather small (e.g., Downs 1957). Drawing on the work of Verba et al. (1995), Kam (2012) theorizes that political participation is higher among those who enjoy taking risks, specifically, because they enjoy new experiences and find political participation stimulating. She finds that risk-accepting individuals are no more likely to vote than those who are risk-averse, but those prone to accepting risks are more likely to engage in nearly all other political participation behaviors, such as attending a campaign rally or contacting an elected official (Kam 2012).[8]

The focus on Kam (2012) is relevant to the present study for two reasons. First, the paper utilizes a scale for measuring risk (shown in Table 1) that includes items culled from prominent surveys such as the General Social Survey. Unfortunately, this includes many items which are inherently abstract in nature—a problem that ought to be significantly helped by the use of aphorisms as measurement tools. For example, several items in Kam's Risk Acceptance scale (KRAS;[9] Kam 2012) ask respondents to engage in broad generalizations of their own attitudes (e.g., item 4: "I like to do frightening things"). Another item asks respondents to place themselves in a hypothetical, abstract situation that they likely

[8]Though Kam (2012) uses reports of both past and future political behavior as dependent variables, I rely solely on past political behaviors due to survey constraints and to avoid asking survey respondents to engage in hypotheticals.

[9]I abbreviate Kam's Risk Acceptance scale to KRAS for ease of reference, submission length, and to distinguish it from Zaller's RAS model.

have never encountered and never considered (item 2, which asks about betting on horses).[10] Such strategies are common in survey research, even though measurements in survey methodology are often more accurate when they include less abstract measures (Krosnick and Presser 2010). For these reasons, I follow a similar process as Kam (2012) in terms of testing a novel risk scale but with a different purpose in mind—validating the use of aphorisms as measures of latent predispositions. Due to potential problems arising from acquiescence bias in prior work (Wilson 2025), I primarily rely on the forced-choice format for measuring latent attitudes rather than standard Likert scales, though I explore both methods.

TABLE 1. Items in Kam's Risk Acceptance Scale (KRAS)

Item	Question wording	Scale
1	Some people say you should be cautious about making major changes in life. Suppose these people are located at 1. Others say that you will never achieve much in life unless you act boldly. Suppose these people are located at 7. And others have views in between. Where would you place yourself on this scale?	7-point scale
2	Suppose you were betting on horses and were a big winner in the third or fourth race. Would you be more likely to continue playing or take your winnings?	5-point scale
3	I would like to explore strange places.	5-point scale
4	I like to do frightening things.	5-point scale
5	I like new and exciting experiences, even if I have to break the rules.	5-point scale
6	I prefer friends who are exciting and unpredictable.	5-point scale
7	In general, how easy or difficult is it for you to accept taking risks?	4-point scale

Note: Items are from Kam's Risk Acceptance Scale (Kam 2012, p. 819).

FORCED-CHOICE FORMAT

Aphorisms are by nature succinct distillations of approaches to life. They are often socialized in situations where there is a direct application of the underlying value. For example, the aphorism "Treat others the way you want to be treated," known in some contexts as the Golden Rule, is commonly seen

[10] A full list of the items contained in KRAS is located in SI Section 1.

across cultures and moral traditions.[11] It is typically applied in situations where an authority figure (e.g., a parent or teacher) is explaining the responsibility to reciprocate desirable social behavior. It is through this and other similar socialization processes that aphorisms become ingrained in cognition regarding one's outlook on life. Through individuals' learned experiences with this saying and others, aphorisms often develop an air of wisdom about them that allows them to be easily assimilated into one's value system.

One of the primary issues that arises with using aphorisms as measurement tools, then, is acquiescence bias. If aphorisms maintain a status of revealing deep truths about the world, survey respondents may be reticent to acknowledge that they disagree with the underlying values conveyed by aphorisms. For example, if the Golden Rule were presented to respondents, it is unlikely many would say they disagree or do not consider it an important precept by which they live their lives. Yet the pitfalls in adopting the Golden Rule as a universal maxim are quite obvious—its moral utility depends on the person adopting it. Those who take pleasure in their own pain or who are suicidal, for example, generally should not treat others the way they would like to be treated.

An approach shown to mitigate the problem of acquiescence bias in survey methodology is the forced-choice format for questions (e.g., Kreitchmann et al. 2019). A prominent example of this in political science is the ANES measure of authoritarianism, or authority-mindedness, which is a battery of four items that asks respondents to choose the qualities they would most like children to possess. Because there are many qualities respondents likely think a child ought to have, this battery of survey questions uses the forced-choice format. Rather than simply agreeing that children should be both obedient and self-reliant, the forced-choice format asks respondents to choose which they find more desirable.

I explore whether this approach might mitigate acquiescence bias in measuring orientations toward aphorisms by using a similar design here. The forced-choice format ought to nudge respondents to think about the differences in worldviews represented by each pair of aphorisms. Respondents might agree with both aphorisms, but the forced-choice approach allows for a more precise estimation of

[11]Though it has presented itself in slightly different forms, Confucius and Aristotle both taught the ethics underlying the Golden Rule, and its use as a maxim has appeared across the world's major religions—Islam, Christianity, Buddhism, Judaism, and Hinduism all contain near-exact or very similar precepts in their teachings.

which underlying worldview conveyed by each aphorism is more relevant to their own outlook on life.

DATA AND METHODS

To test the theory of aphorisms as survey measurement tools, I use survey data collected through a nonprobability national sample of 2,374 respondents recruited by Qualtrics. The survey was conducted in the summer of 2025, and quotas for the sample were matched to Census benchmarks for age, race, sex, and education. Respondents were excluded if they completed the survey 50% faster than the median time for survey duration (see Greszki et al. 2015 for discussion) or if they failed the attention check.

To decide which aphorisms to include in the questionnaire, an initial review of risk-based aphorisms produced around 30 viable candidates to be included in the final scale.[12] After a round of cognitive interviewing and pre-testing, an 11-pair scale was constructed to tap risk-accepting and risk-averse attitudes. The pairs for the Forced-Choice Aphoristic Risk Scale (FCARS) are shown in Table 2. Each aphorism displayed was administered as a single item on a Likert Aphoristic Risk Scale (LARS) for comparison. Due to survey constraints and to reduce repetitiveness of the survey, respondents were randomly assigned to see either the FCARS or the LARS items. Each respondent saw all KRAS items for later comparisons.[13]

The FCARS items began with the following preamble:[14]

[12]This included surveying a database of over 1,000 common aphorisms that was constructed by a team of 12 undergraduate research assistants in the Wilson Research Lab at UNC-Chapel Hill.

[13]Though all respondents saw the KRAS items, question block order of KRAS items and either FCARS or LARS items was randomized as well.

[14]The LARS items began with an amended preamble: *There are many sayings that people use to guide the decisions they make in life, such as "When life gives you lemons, make lemonade." Please read the following statements below and indicate how important, if at all, each of them is to how you approach your own life.*

TABLE 2. Aphorism Pairs for the Forced-Choice Aphorism Risk Scale

Risk-Accepting	Risk-Averse
Fortune favors the bold.	Always look before you leap.
Nothing ventured, nothing gained.	A penny saved is a penny earned.
Don’t put off until tomorrow what can be done today.	All good things come to those who wait.
If at first you don’t succeed, try, try again.	Don’t bite off more than you can chew.
You only live once.	Safety first.
No pain, no gain.	Better safe than sorry.
Live in the moment, not in the future.	Better to have it and not need it than need it and not have it.
You miss 100% of the shots you don’t take.	It’s best to do things right the first time.
Don’t knock it until you try it.	Curiosity killed the cat.
If you shoot for the stars, you just might hit the moon.	A bird in the hand is worth two in the bush.
Life is about the journey, not the destination.	Don’t count your chickens before they hatch.

Note: This table displays FCARS risk-accepting and risk-averse aphorisms. Each aphorism here was a single item on the LARS scale.

There are many sayings that people use to guide the decisions they make in life, such as “When life gives you lemons, make lemonade.” Although there are many sayings like this that most people would agree with, some are more relevant to how they approach life than others. Below, we are going to present you with two common sayings. While both sayings might apply to your own life, please pick the one that is more relevant to how you see the world.

FCARS respondents were then shown each pair of aphorisms and asked to choose which described their outlook on life best. For the LARS items, respondents were given a 4-point scale ranging from “Not important at all” to “Very important.” Because answering 22 items is cognitively burdensome, each respondent in the LARS randomization group was further randomized to see only 12 out of 22 items. Constructions of the LARS scale for later analyses are variations of this subset.

RESULTS

To assess the validity of these aphorisms as measurement tools, I conducted a confirmatory factor analysis to ensure the items were measuring the same latent attitude (i.e., unidimensional). Confirmatory factor analysis provides leverage on this question and ensures that each item loads onto the same underlying construct while also providing substantial additional information about the latent attitude. The common criteria for evaluating confirmatory factor analyses are the following: the standardized root mean square residual (SRMR) below 0.08; the root mean square error of approximation (RMSEA) below 0.05, with a 90% confidence interval; and the comparative fit index (CFI) above 0.95 (Engelhardt et al. 2021; Goretzko et al. 2024; Hetherington et al. 2026).

Because this is an initial analysis of what aphoristic scale might be best to measure risk attitudes, I conducted a series of factor analyses with the FCARS and LARS items to reach a final construction that was optimal. The full scales' factor analyses produced results that were suboptimal (see SI Sections 2 and 3 for full results). An iterative process of removing items that tended to perform relatively poorly (factor loading < 0.4) resulted in final scale constructions that included five FCARS and 15 LARS items, respectively.

TABLE 3. Factor Loadings for FCARS: Final Five-Item Scale

Item	Estimate	SE
Fortune favors the bold	0.573	0.050
Nothing ventured, nothing gained	0.488	0.045
You only live once	0.644	0.042
No pain, no gain	0.843	0.041
You miss 100% of the shots you don't take	0.450	0.045
CFI	0.992	
SRMR	0.033	
RMSEA [90% Confidence Interval]	0.029 [0.000, 0.056]	

Note: Entries in the Estimate column are standardized factor loadings for FCARS items. Model fit statistics are reported at the bottom of the table.

Tables 3, 4, and 5 display the results of the confirmatory factor analyses for the FCARS, LARS,

TABLE 4. Factor Loadings for LARS Items

Item	Estimate	SE
Fortune favors the bold	0.568	0.013
Always look before you leap	–0.626	0.013
A penny saved is a penny earned	–0.601	0.012
Nothing ventured, nothing gained	0.645	0.011
Don't put off until tomorrow	0.635	0.012
All good things come to those who wait	–0.558	0.015
If at first you don't succeed, try again	0.614	0.012
Don't bite off more than you can chew	–0.657	0.011
You only live once	0.560	0.013
No pain, no gain	0.610	0.012
Better safe than sorry	–0.536	0.013
Don't knock it until you try it	0.637	0.010
If you shoot for the moon	0.602	0.012
A bird in the hand is worth two in the bush	–0.571	0.013
Don't count your chickens before they hatch	–0.631	0.014
CFI	0.875	
SRMR	0.076	
RMSEA [90% Confidence Interval]	0.046 [0.041, 0.052]	

Note: Entries in the Estimate column are standardized factor loadings for the LARS items. Negative estimates denote reverse-coded items. Model fit statistics are reported at the bottom of the table.

TABLE 5. Factor Loadings for KRAS Five-Item Scale

Item	Estimate	SE
Act boldly	0.284	0.010
Horse-betting	0.245	0.010
Break the rules	0.711	0.009
Exciting and unpredictable friends	0.627	0.009
Taking risks is easy	0.605	0.010
CFI	0.974	
SRMR	0.026	
RMSEA [90% Confidence Interval]	0.051 [0.039, 0.065]	

Note: Entries in the Estimate column are standardized factor loadings. KRAS items were recoded from Likert responses to a binary agree–disagree scale, with neutral responses coded as 0.5. Model fit statistics are reported at the bottom of the table.

and KRAS items, respectively. All questions were coded on a 0-1 scale such that higher values indicate more risk-acceptance.[15] The five FCARS items appear to load onto a single dimension very well (CFI > 0.95; SRMR < 0.08; and RMSEA < 0.05 [90% CI]), with each item reporting a substantively high factor loading. The LARS items shown in Table 4 appear to somewhat load on the same dimension, with the CFI and RMSEA reporting model fit statistics that are acceptable but not ideal. Reverse-coded LARS items (tapping risk-aversion) appropriately have a negative sign. The KRAS items shown in Table 5 appear to capture the underlying risk dimension well, though the RMSEA is a little high and factor loadings are low for the items tapping respondents' propensity to "act boldly" and continue gambling in a hypothetical horse-betting scenario. In sum, the factor analyses results show that the FCARS items perform better than the LARS items, as expected. Additionally, the FCARS and the KRAS items fit the underlying dimension relatively well.

While these results give an initial idea of the usefulness of each scale, the true utility of attitudinal survey questions lies in the predictive purchase they gain over key variables of interest. Following Kam (2012), respondents were asked whether they engaged in the following forms of political participation in the last 12 months: contacting a federal, state, or local elected official and interacting with elected

[15]Individual FCARS and KRAS items are coded as binaries for ease of comparison between the two scales and then aggregated to create the respective 0-1 scales.

officials on social media.[16] Respondents were also asked if they volunteered for or donated to a political campaign in the 2022 and 2024 elections. If the respective risk scales tap risk-acceptance, those who score higher ought to be more likely to engage in these forms of political participation.

Because aphorisms ought to be most useful to those familiar with them, I restrict the sample to those who were born in the United States. Unfortunately, in the 2025 Qualtrics survey, there was no measure of familiarity with the aphorisms presented due to the added cognitive burden it would place on respondents. A suitable proxy that was asked on the survey is whether the respondent immigrated to the United States or not. Though not perfect, those born in the United States are more likely to be familiar with the American English language customs at an early age, which is crucial for figurative, idiomatic, and proverbial language development (Hartshorne et al. 2018; Levorato and Cacciari 1992; Nippold et al. 1998). I therefore exclude immigrants (N = 208) from the sample and present results for the remaining, non-immigrant sample (N = 2,166).

Table 6 reports the mean levels of risk-acceptance by each measure of political participation. The KRAS and FCARS items each show higher levels of risk among those who engaged in political participation. However, the FCARS items do so more consistently. While the LARS items show higher levels of risk-acceptance among those engaged in political participation compared to those who did not, the differences are often negligible. Given the issues of acquiescence bias with Likert scales, which are likely exacerbated when using aphorisms, this is not surprising.

Table 7 shows the results of linear models that regress each participation measure on each risk scale for non-immigrants. All variables are coded on a 0-1 interval for ease of interpretability. Coefficients represent results from 12 *separate* regressions; Table 7 simply compiles the relevant coefficients for brevity. Each regression controls for gender, race, education, age, income, partisanship, ideology, and political knowledge. Results for the full models with controls can be found in SI Section 4. As with the reported means, the results for the LARS items appear to perform poorly, as expected. The KRAS and FCARS each maintain similar predictive purchase, with each achieving statistical significance on three out of four of the participation measures.

[16]The latter question is a departure from Kam's analysis, but it is in line with the overarching theory. Moreover, risk-acceptance underlies individuals' propensity to post on social media (Bor and Petersen 2022).

TABLE 6. Mean Risk Scale Scores by Political Participation

Participation measure	Response	KRAS	FCARS	LARS
Contacted official	No	0.367	0.304	0.548
		(0.006)	(0.010)	(0.005)
	Yes	0.378	0.370***	0.564
		(0.012)	(0.020)	(0.011)
Interacted on social media	No	0.360	0.307	0.550
		(0.005)	(0.010)	(0.005)
	Yes	0.416***	0.376**	0.561
		(0.015)	(0.023)	(0.012)
Campaign volunteer	No	0.362	0.316	0.551
		(0.005)	(0.009)	(0.005)
	Yes	0.496***	0.380*	0.558
		(0.027)	(0.043)	(0.017)
Campaign donation	No	0.362	0.305	0.545
		(0.005)	(0.010)	(0.005)
	Yes	0.402**	0.380**	0.582**
		(0.014)	(0.022)	(0.012)

Note: Entries are average levels of risk-acceptance by scale and political participation. Standard errors in parentheses. Differences in means that are statistically significant are in bold. ***p < 0.01; **p < 0.05; *p < 0.10

TABLE 7. Risk Orientation Scales and Political Participation

Scale	Contacted official	Interacted on social media	Campaign volunteer	Campaign donation
KRAS	0.065	0.149***	0.084***	0.147***
	(0.041)	(0.038)	(0.023)	(0.038)
FCARS	0.121**	0.109**	−0.004	0.129***
	(0.053)	(0.047)	(0.028)	(0.047)
LARS	0.038	0.024	−0.059	0.165*
	(0.095)	(0.089)	(0.058)	(0.090)

Note: Compiled yet separate OLS regression estimates of risk scales' correlation with each measure of political participation, with standard errors in parentheses. All variables are coded on a 0-1 interval. Controls included but not shown: gender, race, age, income, education, partisanship, ideology, and political knowledge. ***p < 0.01; **p < 0.05; *p < 0.1

DISCUSSION

These findings extend past work (Wilson 2025) by showing that aphorisms are indeed accurate measures of latent non-political attitudes that nonetheless map onto politics. This is the first study that uses an index of aphorisms alone to tap these attitudes, and doing so produced a scale (FCARS) that maps extremely well onto a single latent dimension. As expected, the forced-choice format is likely ideal when measuring respondents' orientations toward aphorisms, unlike standard Likert scales (LARS).

Furthermore, the FCARS items stand up to empirical scrutiny. Past research has shown that those who are more likely to accept risks in their everyday life ought to participate in politics at a higher rate (Kam 2012), and this result was replicated with the FCARS items. When examining the differences in scale means in Table 6, the FCARS items appear to measures risk attitudes' relationship to political participation the most consistently of the three scales tested. Even when controlling for a host of demographic and political factors in Table 7, the effects of risk-acceptance, as measured with the FCARS items, are substantively large and consistent across the four participation measures analyzed, with the exception of volunteering for a campaign.

The theory laid out here suggests that aphorisms themselves are useful as survey measurement tools. Self-reports constructed at the time a survey question is administered can be inaccurate, through no fault of the respondent. For general or abstract survey measurements (e.g., the KRAS items), even respondents engaging with the survey in good faith must read and interpret the question quickly, reflect on their own experiences, and average over these experiences to summarize their views into a category on a Likert scale. Aphorisms asked through the forced-choice format (FCARS items), on the other hand, ought not require the same cognitive burden because respondents are presented with successive pairs of ostensibly familiar aphorisms. Rather than construct an opinion off of the top of their head, respondents are able to lean on their preconceived notions of how they feel about each aphorism as it pertains to their personal approach to life. In other words, survey responses to the FCARS items likely originate in the gut rather than the head.

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Supporting Information Table of Contents

1 KAM'S RISK ACCEPTANCE SCALE

Items contained in Kam's Risk Acceptance Scale (Kam, 2012, p. 819):

1. Some people say you should be cautious about making major changes in life. Suppose these people are located at 1. Others say that you will never achieve much in life unless you act boldly. Suppose these people are located at 7. And others have views in between. Where would you place yourself on this scale?
2. Suppose you were betting on horses and were a big winner in the third or fourth race. Would you be more likely to continue playing or take your winnings? (Definitely continue playing; Probably continue playing; Not sure; Probably take my winnings; Definitely take my winnings)
3. I would like to explore strange places. (Strongly disagree to strongly agree, 5 categories)
4. I like to do frightening things. (Strongly disagree to strongly agree, 5 categories)
5. I like new and exciting experiences, even if I have to break the rules. (Strongly disagree to strongly agree, 5 categories)
6. I prefer friends who are exciting and unpredictable. (Strongly disagree to strongly agree, 5 categories)
7. In general, how easy or difficult is it for you to accept taking risks? (Very easy; Somewhat easy; Somewhat difficult; Very difficult)

2 FCARS FULL-SCALE FACTOR ANALYSIS

TABLE 8. Factor Loadings for FCARS: All 11 Items

Item	Estimate	SE
Fortune favors the bold	0.555	0.050
Nothing ventured, nothing gained	0.539	0.041
Don't put off until tomorrow	0.168	0.046
If at first you don't succeed, try again	0.304	0.047
You only live once	0.631	0.038
No pain, no gain	0.756	0.035
Live in the moment	0.285	0.049
You miss 100% of the shots you don't take	0.480	0.041
Don't knock it until you try it	0.300	0.047
Shoot for the moon	0.414	0.043
Life is about the journey	0.374	0.043
CFI	0.887	
SRMR	0.074	
RMSEA [90% Confidence Interval]	0.049 [0.040, 0.057]	

Note: Entries in the Estimate column are standardized factor loadings for FCARS items. Model fit statistics are reported at the bottom of the table.

TABLE 9. Factor Loadings for FCARS: Items with Previous Loadings Above 0.4

Item	Estimate	SE
Fortune favors the bold	0.573	0.050
Nothing ventured, nothing gained	0.504	0.043
You only live once	0.643	0.041
No pain, no gain	0.821	0.038
You miss 100% of the shots you don't take	0.468	0.043
Shoot for the moon	0.379	0.045
CFI	0.990	
SRMR	0.035	
RMSEA [90% Confidence Interval]	0.026 [0.000, 0.047]	

Note: Entries in the Estimate column are standardized factor loadings for FCARS items. Model fit statistics are reported at the bottom of the table.

3 LARS FACTOR ANALYSES

TABLE 10. Factor Loadings for LARS: All 22 Items

Item	Estimate	SE
Fortune favors the bold	0.572	0.012
Always look before you leap	–0.627	0.013
A penny saved is a penny earned	–0.616	0.011
Nothing ventured, nothing gained	0.627	0.010
Don't put off until tomorrow	0.652	0.011
All good things come to those who wait	–0.551	0.014
If at first you don't succeed, try again	0.610	0.012
Don't bite off more than you can chew	–0.648	0.010
Safety first	–0.542	0.012
You only live once	0.559	0.012
No pain, no gain	0.594	0.011
Better safe than sorry	–0.559	0.012
Better to have it and not need it	–0.539	0.012
Live in the moment	0.409	0.014
You miss 100% of the shots you don't take	0.488	0.014
It's best to be prepared	–0.459	0.013
Curiosity killed the cat	–0.518	0.013
Don't knock it until you try it	0.628	0.010
If you shoot for the moon	0.613	0.012
A bird in the hand is worth two in the bush	–0.578	0.013
Life is about the journey, not the destination	0.538	0.012
Don't count your chickens before they hatch	–0.637	0.013
CFI	0.849	
SRMR	0.076	
RMSEA [90% Confidence Interval]	0.042 [0.038, 0.045]	

Note: Entries in the Estimate column are standardized factor loadings for the LARS items. Negative estimates denote reverse-coded items. Model fit statistics are reported at the bottom of the table.

4 FULL REGRESSION MODELS

TABLE 11. Full Models Predicting Contacting an Official

	KRAS	FCARS	LARS
KRAS	0.065		
	(0.041)		
FCARS		0.121*	
		(0.053)	
LARS			0.038
			(0.095)
Female	0.012	0.005	−0.001
	(0.019)	(0.030)	(0.028)
Black	−0.047	−0.043	−0.053
	(0.033)	(0.052)	(0.050)
Asian	−0.067	−0.200	0.101
	(0.112)	(0.151)	(0.180)
Indigenous	0.055	−0.073	0.197
	(0.103)	(0.163)	(0.146)
Other/mixed/NA	0.008	−0.009	0.036
	(0.049)	(0.070)	(0.078)
Hispanic	0.021	0.003	0.045
	(0.041)	(0.060)	(0.063)
High school or less	−0.201***	−0.224***	−0.187***
	(0.035)	(0.052)	(0.051)
Some college	−0.149***	−0.161**	−0.147**
	(0.034)	(0.051)	(0.050)
College	−0.123***	−0.105*	−0.143**
	(0.035)	(0.052)	(0.051)
Age	0.009	0.073	−0.006
	(0.040)	(0.063)	(0.059)
Income	0.053	0.006	0.122*
	(0.034)	(0.052)	(0.050)
Republican	0.066	0.044	0.090
	(0.050)	(0.090)	(0.076)
Democrat	0.080	0.079	0.078
	(0.049)	(0.090)	(0.074)
Ideology	−0.135**	−0.136*	−0.153*
	(0.042)	(0.065)	(0.063)
Political knowledge	0.151***	0.081†	0.117*
	(0.029)	(0.047)	(0.047)
Observations	1,874	896	873
Adjusted R^2	0.075	0.063	0.065

TABLE 12. Full Models Predicting Contacting on Social Media

	KRAS	FCARS	LARS
KRAS	0.149***		
	(0.038)		
FCARS		0.109*	
		(0.047)	
LARS			0.024
			(0.089)
Female	0.037*	0.034	0.018
	(0.018)	(0.027)	(0.026)
Black	0.060*	0.101*	0.060
	(0.030)	(0.047)	(0.047)
Asian	−0.039	−0.051	−0.036
	(0.102)	(0.135)	(0.169)
Indigenous	0.105	−0.021	0.245†
	(0.093)	(0.146)	(0.137)
Other/mixed/NA	−0.009	−0.006	−0.011
	(0.044)	(0.063)	(0.073)
Hispanic	0.052	0.042	0.081
	(0.038)	(0.054)	(0.059)
High school or less	−0.133***	−0.134**	−0.139**
	(0.031)	(0.047)	(0.048)
Some college	−0.098**	−0.099*	−0.099*
	(0.031)	(0.046)	(0.047)
College	−0.100**	−0.073	−0.134**
	(0.032)	(0.047)	(0.048)
Age	−0.027	0.016	−0.071
	(0.037)	(0.057)	(0.056)
Income	−0.003	−0.003	0.028
	(0.031)	(0.047)	(0.047)
Republican	0.120**	0.141†	0.157*
	(0.045)	(0.081)	(0.071)
Democrat	0.111*	0.146†	0.121†
	(0.044)	(0.081)	(0.069)
Ideology	−0.055	−0.028	−0.099†
	(0.038)	(0.058)	(0.059)
Political knowledge	0.118***	0.044	0.104*
	(0.026)	(0.042)	(0.044)
Observations	1,874	896	873
Adjusted R^2	0.041	0.021	0.028

TABLE 13. Full Models Predicting Campaign Volunteering

	KRAS	FCARS	LARS
KRAS	0.084***		
	(0.023)		
FCARS		−0.004	
		(0.028)	
LARS			−0.059
			(0.058)
Female	−0.003	−0.016	−0.014
	(0.011)	(0.016)	(0.017)
Black	0.043*	0.022	0.096**
	(0.019)	(0.027)	(0.030)
Asian	−0.007	−0.087	0.088
	(0.063)	(0.079)	(0.109)
Indigenous	0.003	−0.010	0.035
	(0.058)	(0.085)	(0.089)
Other/mixed/NA	−0.017	0.032	−0.087†
	(0.028)	(0.037)	(0.047)
Hispanic	−0.021	−0.056†	0.025
	(0.023)	(0.031)	(0.038)
High school or less	−0.055**	−0.055*	−0.066*
	(0.020)	(0.027)	(0.031)
Some college	−0.044*	−0.034	−0.060*
	(0.019)	(0.027)	(0.030)
College	−0.040*	0.008	−0.091**
	(0.020)	(0.028)	(0.031)
Age	−0.115***	−0.079*	−0.182***
	(0.023)	(0.033)	(0.036)
Income	0.026	−0.010	0.089**
	(0.019)	(0.027)	(0.030)
Republican	0.044	0.014	0.074
	(0.028)	(0.047)	(0.046)
Democrat	0.033	0.010	0.041
	(0.028)	(0.047)	(0.045)
Ideology	−0.074**	−0.094**	−0.083*
	(0.024)	(0.034)	(0.038)
Political knowledge	−0.001	−0.013	−0.047†
	(0.016)	(0.024)	(0.028)
Observations	1,874	896	873
Adjusted R^2	0.050	0.029	0.083

TABLE 14. Full Models Predicting Campaign Donation

	KRAS	FCARS	LARS
KRAS	0.147***		
	(0.038)		
FCARS		0.129**	
		(0.047)	
LARS			0.165†
			(0.090)
Female	−0.013	−0.005	−0.057*
	(0.018)	(0.027)	(0.027)
Black	0.021	0.038	0.031
	(0.031)	(0.046)	(0.047)
Asian	0.297**	0.156	0.507**
	(0.103)	(0.135)	(0.171)
Indigenous	0.133	−0.029	0.330*
	(0.094)	(0.146)	(0.139)
Other/mixed/NA	−0.091*	−0.010	−0.208**
	(0.045)	(0.063)	(0.074)
Hispanic	0.082*	0.050	0.153*
	(0.038)	(0.054)	(0.060)
High school or less	−0.119***	−0.087†	−0.151**
	(0.032)	(0.046)	(0.049)
Some college	−0.087**	−0.067	−0.107*
	(0.031)	(0.046)	(0.047)
College	−0.045	−0.002	−0.091†
	(0.032)	(0.047)	(0.048)
Age	0.037	0.093†	−0.008
	(0.037)	(0.056)	(0.056)
Income	0.056†	0.028	0.124**
	(0.031)	(0.046)	(0.047)
Republican	0.066	0.002	0.167*
	(0.046)	(0.081)	(0.072)
Democrat	0.080†	0.036	0.146*
	(0.045)	(0.081)	(0.070)
Ideology	−0.124**	−0.117*	−0.146*
	(0.039)	(0.058)	(0.060)
Political knowledge	0.194***	0.181***	0.142**
	(0.026)	(0.042)	(0.044)
Observations	1,874	896	873
Adjusted R^2	0.097	0.075	0.107